

PROJECT PROPOSAL

*Recall AI – Proactive Study Assistant, Tracing Weak
Knowledge Back to Its Root*

Course: Introduction to Software Engineering

Team: CAGT

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1. Team Information

- **Team name:** CAGT
- **Number of members:** 5

No.	Full Name	Primary Role	Secondary Role
1	Thái Nguyễn Tuấn Kiệt	Project Manager	UI/UX Designer
2	Nguyễn Thế Quân	System Architect	Fullstack Developer
3	Nguyễn Minh Phát	QA/Tester	Fullstack Developer
4	Nguyễn Phương Gia Bảo	Frontend Leader	-
5	Ngô Văn Phong	Backend Leader	-

2. Project Introduction

2.1. Idea Summary

Recall AI is a web app that helps students turn study materials (outlines, slides, notes) into a structured study plan, helping students verify whether they truly understand the material or have only become visually familiar with the text. The system uses AI as an examiner, asking multi-turn oral-exam-style questions based strictly on the material the user uploaded, while simultaneously building a prerequisite relationship graph among the concepts in that material.

When a student answers incorrectly, the system does not simply reschedule that exact question a few days later like most existing review tools - it traces back through the graph to check whether the student has a gap in an underlying foundational concept, and automatically inserts a review of that foundational concept into the schedule before returning to the original question. This mechanism is detailed in Section 5.4, and is the technical centerpiece of the entire project.

2.2. Problem & Motivation

- **Passive learning, shallow review:** Most students review by re-reading or highlighting - a method with low retention effectiveness compared to retrieval practice, according to cognitive psychology research.
- **No objective way to self-assess:** Students find it hard to know whether they have truly “mastered” material or merely “recognize the words,” since no one checks their real level of understanding.
- **Unplanned review:** Before deadlines or exams, students often cram, without breaking content down by priority or remaining time.

These three problems are tightly linked. Today’s AI can address two parts at once: reading comprehension of documents to build a plan, and acting as an examiner to test knowledge - something only humans could previously do. However, as discussed in Section 2.3, “AI quizzing based on user-uploaded material” has already been implemented extremely well by many commercial tools. The gap the team has identified - and therefore the technical focus of the project - is that no existing tool traces an incorrect answer back to the exact foundational concept that caused it, in order to review the true root cause rather than repeating the same symptom.

2.3. Competitive Analysis

Before finalizing the technical direction, the team surveyed widely used AI-based review tools, current as of June 2026: Google NotebookLM, Quizlet (Q-Chat, Learn Mode), RemNote, and the option of a self-built Custom GPT/Claude Project. Survey results:

Criteria	NotebookLM	Quizlet	RemNote	Custom GPT/Claude Project	Recall AI
Generates questions/cards from the user’s own material	Yes	Yes (Magic Notes)	Yes	Yes, but manual re-pasting each time	Yes

Criteria	NotebookLM	Quizlet	RemNote	Custom GPT/Claude Project	Recall AI
Spaced repetition	Partial (filters “previously wrong” answers, not a full spacing algorithm)	Yes, mature since 2017 (Learn Mode)	Yes, 3-axis model: recall, stability, difficulty	None, must reopen manually	Yes - required, but not the main differentiator
Application/transfer questions, not just definitions	Requires custom prompts	Built-in (“Apply My Knowledge”)	None	Requires knowing prompts	Built-in by default
Multi-turn, stateful conversational quizzing	Very new (Learning Guide, 07/2026), depth unclear	Yes, optional mode users must enable	None (notes + cards only)	Yes, but user manages the flow	Yes, by default, clear turn limits
Traces back to prerequisite concepts on wrong answers	No	No	No	No	Yes
Optimized for Vietnamese &	Not a focus	Not a focus	Not a focus	Depends on model	Narrow niche, better

Criteria	NotebookLM	Quizlet	RemNote	Custom GPT/Claude Project	Recall AI
Vietnamese engineering terms					investment possible

On 5 of the 6 comparison criteria, existing tools - especially NotebookLM, Quizlet, and RemNote - have already implemented these features well, with resources and user data far beyond what a team of 5 students could achieve in 10 weeks. The team therefore does not consider these criteria to be Recall AI's differentiators, although they must still be implemented well enough for the product to be usable. The only criterion that no tool has yet implemented for freely user-uploaded material is tracing back through prerequisite relationships between concepts when a weakness is detected - this is chosen as the project's technical focus, detailed in Section 5.4.

Tracing back through prerequisite relationships is not an entirely new idea in education science. Knowledge Space Theory - applied in the ALEKS platform for over 25 years - uses this exact principle to model a skill graph with prerequisite relationships for each student. Recall AI's difference is applying this principle to any material a user uploads, through automatic AI extraction, rather than requiring an expert-designed curriculum like ALEKS, which is limited to math, chemistry, and statistics.

2.4. Desired Outcomes

- Students upload study materials and receive a structured study plan, arranged by deadline, within minutes.
- Through multi-turn quizzing, the system distinguishes rote-memorized answers from answers reflecting genuine understanding, based on concrete evidence in the conversation.
- When a weak concept is detected, the system automatically identifies and suggests reviewing the exact foundational concept causing that weakness, without waiting for the student to realize it themselves.

- By the end of the review period, students have a visual picture (a concept graph color-coded by mastery level) of the entire subject, instead of a disconnected list of scores.
- The team demonstrates the ability to design and build a system with real orchestration logic (scheduling algorithms, graph traversal) behind the AI-calling interface layer.

3. Target Users

First- and second-year students in engineering/natural science majors, especially in subjects with a heavy theoretical load and a clearly layered concept structure - meaning that understanding one concept usually requires having mastered a prior one (e.g., computer networks, operating systems, data structures & algorithms, foundational courses). This “layered” characteristic is the condition for the prerequisite-tracing feature (Section 5.4) to add value; for subjects with discrete, independent content, the system still works but operates more like an ordinary AI-based review tool.

4. Core Value & Experience Loop

1. **Ingest & Map:** AI reads the study material (text or photographed images), extracting a list of concepts and the prerequisite relationships between them. The user reviews and can edit the graph before starting to study.
2. **Schedule:** The system arranges concepts into small study sessions, prioritized by the user-provided deadline and estimated difficulty.
3. **Verify (Interview):** After each study session (with or without Pomodoro), AI acts as an examiner, conducting a multi-turn oral quiz based on the uploaded material, within a pre-configured turn limit, and records an understanding score for each concept.
4. **Remediate:** When a concept’s score falls below a threshold, the system traces back through the prerequisite relationship graph to check related foundational concepts, and automatically inserts a review of that concept into the schedule before re-asking the original concept.

5. **Reflect:** The Dashboard visually displays the concept graph, color-coded by mastery level, helping students immediately see the overall picture of the subject.

5. Key Features

5.1. AI Study Planner

- Input: text (pasted outline/notes) or a photo of the material, processed directly via LLM.
- AI returns a fixed-structure JSON result: list of concepts, prerequisite relationships between them, estimated difficulty - not free-form text, so the system can use it directly for the scheduling step without needing to reinterpret it.
- If the AI returns an incorrect format, the system retries or splits the material by heading/paragraph so the entire feature doesn't fail.
- Users can edit and manually reorder items if desired.

5.2. Focus Session

- Users can choose a study method (default: Pomodoro).
- Session history is saved to display study time statistics.

5.3. AI Examiner – Multi-turn Interview

The AI Examiner is the feature that directly operationalizes the product's knowledge-verification mechanism, distinct from isolated, independent question-answer pairs:

- **Multi-turn, stateful conversation with clear limits:** Based on the user's answer, the system decides the next step within a bounded state machine: a maximum of N question-answer turns per concept (e.g., 3 turns), to control AI call costs and response time.
- An answer showing genuine understanding → the system asks a deeper follow-up within the remaining turn limit (e.g., "You just said X, so why does Y happen?").
- A shallow/rote-memorized answer → the AI probes further to check real understanding, not accepting generic answers.

- A missing/incorrect answer → the AI points out the mistake; if it relates to a foundational concept, the system switches to the trace-back mechanism in Section 5.4 instead of letting the AI decide on its own which concept to re-ask.
- **Separating the AI's role from the orchestration role:** The AI is only called for two narrow tasks with fixed output schemas - generating a question (*generate_question*) and grading an answer against a rubric extracted from the original material (*grade_answer*). Decisions about when to stop or when to switch concepts are deterministic software logic, making system behavior testable and independent of whether the AI responds consistently across calls.
- **Firmly anchored to the source material:** All questions and grading criteria are based strictly on the content the user uploaded.
- **End-of-session summary comment:** Synthesized from the scores saved for each concept, then the AI only rewrites it into a natural-language comment - what's solid, what's still weak, suggestions for what to review - without fabricating an assessment from conversational memory.
- **Fallback mechanism when AI fails/runs out of quota:** The system switches to a self-grading mode using previously generated questions (as static flashcards); the user self-assesses correct/incorrect, and the system still updates the review schedule normally.
- **Voice Input:** Adding Speech-to-Text may be considered at a later stage; not included in the MVP scope.

5.4. Concept Graph Engine

This is the component receiving the most technical investment in the project, and is the basis for Recall AI's differentiation from the tools surveyed in Section 2.3.

Mechanism

6. **Extraction:** At the Ingest step, in addition to the list of concepts, the AI is asked to also return the prerequisite relationships between concepts, in the form `{concept_id, prerequisite_of:[concept_id, ...]}`. The system checks that this output forms a directed acyclic graph (DAG) before saving; if a cyclic relationship is detected,

the system requests regeneration or removes the edge causing the cycle and logs it for review.

7. **Manual confirmation:** The user views the graph as a visual diagram and can edit, delete, or add relationships before starting to study. The system does not fully trust the AI-generated result automatically.
8. **Storage:** The graph is stored in PostgreSQL using two relational tables, without needing a dedicated graph database: `concepts (id, plan_id, name, mastery_score, last_tested_at);concept_edges (from_concept_id, to_concept_id)` - where “from” is a prerequisite of “to”.
9. **Trace-back algorithm**, run after every grading turn in an Interview session:

```
WHEN concept C has mastery_score < threshold (e.g., 0.6):
```

```
  Q = empty queue
```

```
  ADD the direct prerequisites of C (from concept_edges) to Q
```

```
  TRAVERSE Q, limited to a maximum of 2 levels to avoid overly long sessions:
```

```
    IF prerequisite P has never been tested OR mastery_score(P) < threshold:
```

```
      INSERT P at the front of the next session's queue, ahead of C
```

```
    ELSE:
```

```
      skip P (already solid, no need to review)
```

```
  IF no prerequisite needing further review is found:
```

```
    handle as ordinary spaced repetition - schedule C for review again after X days
```

This is a reverse breadth-first search (BFS) algorithm on the graph, with a depth limit to avoid excessively long sessions. The algorithm is independent of AI quality - it can be unit tested using mock `mastery_score` data and mock relationships, with no need to call the actual AI.

Limitation to note: this is a simplified version, inspired by Knowledge Space Theory, not a full implementation of a probabilistic model like ALEKS (Section 2.3). The quality of this mechanism depends on the AI correctly extracting prerequisite relationships, so the manual confirmation step above is mandatory, not optional.

5.5. Dashboard

- Review progress per subject/material, upcoming deadlines.
- Concept graph diagram, color-coded by mastery level based on **mastery_score** - reflecting real understanding level and the position of weaknesses within the knowledge chain, not just a disconnected list of topics.

5.6. Account Management

- Sign in/sign up with email + password (or Google account), basic profile page.

6. Agentic Elements

- **Graph-based trace-back and session routing:** The system automatically inserts prerequisite concepts into the schedule when it detects that a weakness originates from a root cause, without waiting for the user to realize it or request it themselves (Section 5.4). This is agentic behavior that can be pointed to directly in the algorithm during reporting, not just a system prompt.
- **AI makes narrow-scope decisions at each question-answer turn:** going deeper or switching topics, but always within a fixed turn limit enforced by the software (Section 5.3).
- **Proactive, context-aware reminders:** The system calculates and sends review reminders based on the remaining time until the deadline and the memory retention level measured through the Examiner, rather than reminders on a fixed schedule.

Note: This is the part receiving technical investment in the middle phase of the project (graph schema design, trace-back algorithm), and is also the easiest part to demonstrate when reporting that the product is not just an interface calling an API.

7. User Interviews

7.1. Interview Summary Table

No.	Interviewee Name	Occupation / Background	Interview Date	Format (In-person/Online)	Evidence Link
1	Nguyen Phi Hung	Information Security	23/06/2026	In-person	https://drive.google.com/drive/folders/1yBxeORowbpG5cygIvgIVmAKCs-TQDs6?usp=sharing
2	Ta Minh Dat	Law	23/06/2026	In-person	https://drive.google.com/drive/folders/160CKFUx28THslMKqiWZNtuSHXCCy7nY_?usp=sharing
3	Pham Chi Tam	Electronics and Telecommunications	23/06/2026	In-person	https://drive.google.com/drive/folders/1p1K7OfY1Ugc-pKlxY8LWNWtK_7M1Cz5m?usp=sharing

No.	Interviewee Name	Occupation / Background	Interview Date	Format (In-person/Online)	Evidence Link
4	Ngo Van Phong	Information Technology	23/06/2026	In-person	https://drive.google.com/drive/folders/1IlwJmgO7qRR-DK0s8TNdAnr7xj3FmhRU?usp=sharing
5	Nguyen Minh Phat	Information Technology	23/06/2026	In-person	https://drive.google.com/drive/folders/1bpBban2X98XRft3AptCxPmsZw7AjDRJy?usp=sharing

7.2. Key Findings from Interviews

- **Finding 1:** AI knowledge review after completing a task is widely welcomed, but it must ask specific, targeted questions, and only apply to academic content.
- **Finding 2:** The Pomodoro timer is the lowest-rated feature because in reality, tasks often exceed the estimated time.

7.3. Project Adjustments Based on Interview Feedback

Idea/Change	Source (Which Interview)	Decision (Adopt/Skip)	Reason
AI only generates a flexible skeleton plan for reference, rather than a rigid, hour-by-hour schedule. Users retain full editing control.	Nguyen Phi Hung	Adopt	Detailed schedules create pressure, making users feel controlled by the AI. A reference plan provides enough guidance while remaining flexible for real-life changes.
The Review feature (AI Examiner) is only triggered for academic tasks, not automatically activated for daily activities (e.g., gym, shopping, hanging out).	Nguyen Phi Hung, Pham Chi Tam	Adopt	Users find reviews of personal daily activities annoying and invasive. Reviews should be strictly reserved for knowledge-based content.
The AI Examiner must ask specific, targeted questions directly tied to the source material (e.g., formulas, key concepts, problem-solving approaches), rather than vague questions like "Did	Nguyen Phi Hung	Adopt	Vague questions provide no real value and fail to assess the user's actual comprehension. This also aligns perfectly with the existing "AI

Idea/Change	Source (Which Interview)	Decision (Adopt/Skip)	Reason
you understand the lesson?"			Examiner anchored to source material" design in PA0.
The Focus Timer acts as a soft estimate and does not hard-stop when time runs out. Users manually end the session whenever they are actually finished.	Pham Chi Tam, Nguyen Phi Hung	Adopt	In reality, tasks often exceed their estimated duration. A hard timer forces users to either abandon their work midway or mark it as complete even when it is unfinished.
Consider introducing a feature to block distracting apps (e.g., TikTok, Instagram) during active Focus Sessions.	Nguyen Minh Phat	Skip	App blocking capabilities depend on OS limitations and system permissions, making implementation highly complex. This is currently out of scope for the MVP but can be suggested as a future enhancement.

8. Expected Screen List (MVP System Scope)

N o.	Screen	Short Description	Planned Sprint
1	Landing Page	App introduction page, sign-up CTA	3
2	Sign Up / Sign In	Email + password form	3
3	Dashboard	Overview of active study plans, nearest deadline, concept graph color-coded by mastery level	3 -> graph portion completed in Sprint 5
4	Create Study Plan	Input text/image/PDF/DOCX material, input deadline, view and edit AI-suggested concept graph, submit to AI Study Planner for processing	3
5	Focus Session	Choose a study method (Pomodoro, ...)	3
6	AI Examiner – Interview	Multi-turn conversation interface between user and AI examiner, based on uploaded material	4
7	Interview Session Result	End-of-session summary: solid areas, weak areas, review suggestions	4
8	History & Progress	Concept graph over time, history of Interview and Focus sessions	5
9	Personal Profile	Basic account information, change password	3 (minimal)

9. Tech Stack

9.1. Overview Table

Component	Technology
Frontend	React (Vite) + TypeScript + Tailwind CSS + shadcn/ui + React Router + Axios
Graph visualization	react-flow
Backend	Node.js + Express.js
Database	PostgreSQL + Prisma ORM (including 2 tables for the Concept Graph Engine: concepts , concept_edges)
Authentication	JWT + bcrypt
AI Services	Gemini API - output required to follow a fixed JSON schema (concepts, prerequisites, questions, grading), no free-form text used for logic
Scheduling & Remediation Engine	Custom-written algorithms: priority queue (scheduling) + depth-limited reverse BFS (prerequisite trace-back)
Source Control	Git + GitHub
Project Mgmt	JIRA
Design	Figma, Stitch
Communication	Discord

9.2. Rationale, Advantages and Risks

Frontend: React (Vite) + TypeScript + Tailwind CSS + shadcn/ui

Rationale: Recall AI is a Single Page Application (SPA) that does not need SEO. React combined with Vite is the leanest solution for an SPA. TypeScript helps catch errors strictly, reducing potential bugs. shadcn/ui helps quickly build high-quality UI screens without needing to code complex components from scratch.

Advantages:

- TypeScript ensures type safety, helping Frontend and Backend stay in sync on data structures; IntelliSense speeds up development.
- shadcn/ui provides high-quality pre-built components that are customizable since the component code lives directly in the project.
- Vite provides fast Hot Module Replacement (HMR).
- Tailwind CSS enables fast UI building, integrating well with shadcn/ui.

Risks:

- TypeScript has a certain learning curve for newcomers - mitigated by having the Frontend Leader set up basic types and provide pair-programming guidance.
- Can be overwhelming due to the amount of code generated by shadcn/ui - mitigated by only installing components that are truly necessary.

Backend: [Node.js](#) + [Express.js](#)

Rationale: Express.js is a minimal, flexible framework with the lowest learning curve in the Node.js ecosystem. With a 10-week timeframe and a team of 5, a heavier framework like NestJS would consume time better spent focusing on business logic.

Advantages:

- Same JavaScript/TypeScript language as Frontend, allowing types/interfaces to be shared between the two layers.
- Large community, with common middleware readily available (cors, helmet, morgan, multer...).
- Unopinionated architecture, allowing the team to organize code in the way that best fits them.

Risks:

- No default project structure, which can easily lead to inconsistent code - mitigated by clear folder conventions (routes/, controllers/, services/, middlewares/) and ESLint/Prettier from the start.
- Error handling and validation require manual configuration - mitigated by using a validation library such as Zod.

Database: PostgreSQL + Prisma ORM

Rationale: Recall AI's data has clear relationships (User → Plan → Concept → ConceptEdge → FocusSession → VerifyResult). PostgreSQL ensures better data integrity than NoSQL for this relational model, including representing the concept graph using two relational tables instead of a dedicated graph database. Prisma ORM auto-generates TypeScript types from the schema, catching errors at code time instead of runtime.

Advantages:

- PostgreSQL supports ACID transactions, ensuring data consistency when multiple operations occur simultaneously.
- Prisma Migrate automates schema change management across Sprints.
- Prisma Client generates type-safe queries, reducing runtime errors related to the database.
- Does not introduce a new type of database into the system just to serve the concept graph, keeping the architecture simple - suitable for a team of 5 over 10 weeks.

Risks:

- Prisma has its own schema language (.prisma), requiring time to get used to - mitigated by dedicating time in Sprint 2 to carefully design the schema, including the **concepts/concept_edges** tables.
- Schema changes between Sprints can cause migration conflicts - mitigated by careful design during the Elaboration phase, limiting major changes during Construction.
- For very large graphs, recursive relational queries would be slower than a dedicated graph database; at MVP scale (one study document, typically under 50 concepts), this risk is negligible.

Authentication: JWT + bcrypt

Rationale: Building the authentication system in-house helps the team deeply understand security mechanisms - important knowledge for the course - while avoiding third-party service costs.

Advantages:

- Full control over the authentication flow, easy to customize.

- bcrypt is a proven, secure password-hashing algorithm that automatically adds salt.
- JWT enables stateless authentication, reducing server load.

Risks:

- Requires implementing refresh token rotation, token blacklisting on logout, and protection against CSRF/XSS - mitigated by following established best practices and thoroughly reviewing the authentication code.
- Potential security vulnerabilities if implemented incorrectly - mitigated by prioritizing the highest level of review for the authentication portion.

AI Services: Gemini API (Google)

Rationale: Gemini is a multimodal model (text and image), suited to the AI Study Planner requirement (users may photograph material or enter text). The free tier is sufficient for the development and demo phases.

Advantages:

- Strong Vietnamese language support, important for a product targeting Vietnamese users.
- Simple API, clear documentation, quick to integrate.
- Handles multimodal input (text + image) in a single API call, without needing a separate OCR pipeline.

Risks:

- Dependency on an external service: if the API experiences issues or changes pricing, the application is affected - mitigated by an abstraction layer (AI service layer) allowing a provider switch if needed.
- Rate limits on the free tier may restrict the number of requests - mitigated by caching AI results and limiting requests per user per day.
- Inconsistent output quality, especially with complex input or low-quality images - mitigated by careful prompt design and always enforcing a validatable JSON schema for output.

Concept Graph & Visualization: PostgreSQL (relational tables) + react-flow

Rationale: Prerequisite relationships between concepts can be fully represented using two relational tables (**concepts**, **concept_edges**) in the PostgreSQL database already chosen for the whole system - no need to invest in a separate graph database (e.g., Neo4j), which would add unnecessary operational complexity at MVP scale. For visualizing the graph on the Dashboard, the team uses react-flow - a ready-made React library for node-link interfaces - to focus development time on the trace-back algorithm rather than building visualization from scratch.

Advantages:

- Does not introduce a new type of database, keeping the architecture simple.
- Prisma ORM queries these tables like any other table, with no need for a separate driver.
- react-flow already handles layout, drag-and-drop interaction, and zoom for node-link diagrams.

Risks:

- react-flow has its own learning curve for the Frontend team - mitigated by using only the basic API (nodes, edges, coloring) needed for the MVP.

10. Roles & Responsibilities

10.1. Detailed Assignment Table

Member	Primary Role	Scope of Responsibility
Thái Nguyễn Tuấn Kiệt	Project Manager + UI/UX Designer	Manages progress, drafts specs, designs wireframes/UI, prepares Sprint reports
Nguyễn Thế Quân	System Architect + Fullstack Developer	Designs system architecture, database schema, Concept Graph Engine algorithms, API design, fullstack development

Member	Primary Role	Scope of Responsibility
Nguyễn Minh Phát	QA/Tester + Fullstack Developer	Writes test plans/cases, performs testing, supports backend development
Nguyễn Phương Gia Bảo	Frontend Leader	Leads the frontend team, implements UI components, visualizes the concept graph, integrates APIs
Ngô Văn Phong	Backend Leader	Leads the backend team, implements API endpoints, manages the database

10.2. RACI Matrix (for Key Activities)

Activity	Kiệt (PM)	Quân (Arch)	Phát (QA)	Bảo (FE)	Phong (BE)
Sprint progress management	R/A	C	I	I	I
Architecture design	C	R/A	I	C	C
UI/UX design	R/A	C	C	C	C
Concept Graph Engine design & implementation	C	R/A	C	I	C
Graph visualization on Dashboard	I	C	I	R/A	C
Frontend development	I	C	I	R/A	I
Backend development	I	C	I	I	R/A
AI integration (Gemini API)	I	R	I	C	A
Testing	I	I	R/A	C	C
Reporting & Documentation	R/A	C	C	I	I

R = Responsible, A = Accountable, C = Consulted, I = Informed

11. Project Schedule

Sprint	RUP Phase	Timeline	Assignment	Main Objective	Content
1	Inception	Jun 21 – Jun 28	PA0	Project proposal	Finalize the project proposal (this document).
2	Elaboration	Jun 29 – Jul 12	PA1, PA2	Requirements and initial design	Detailed requirements specification (use cases, user stories), system architecture design, database design (including tables for concepts and prerequisite relationships), on-paper trace-back algorithm design, wireframes for screens. No coding yet.
3	Elaboration	Jul 13 – Jul 26	PA3	Detailed design and test planning. Working software	Detailed design + test plan; build the first working software: Auth, Dashboard skeleton, AI Study Planner (JSON schema, concept and prerequisite extraction), first version of Concept Graph Engine (graph storage, trace-back algorithm working with mock data), Focus Session.
4	Construction	Jul 27 – Aug 9	PA4	Working software. Implementation and testing	AI Examiner – multi-turn interview (turn-limited state machine), connect grading results to the Concept Graph Engine built in Sprint 3, test conversation flow.

Sprint	RUP Phase	Timeline	Assignment	Main Objective	Content
5	Construction	Aug 10 – Aug 23	PA5	Working software. Implementation and testing	Visualize the concept graph on the Dashboard, complete the agentic layer (auto-suggesting inserted review sessions based on the graph), test the entire system with real data, prepare the demo.

The Concept Graph Engine - the project’s highest-risk and highest-value component - is scaffolded starting in Sprint 3 and tested with mock data, before the AI Examiner (Sprint 4) produces real scoring data to connect to it. This sequencing separates the risk of “an incorrect trace-back algorithm” from the risk of “an unpolished AI Examiner,” so the two risks don’t both pile up in the final Sprint.

12. Risks & Assumptions

- **AI Examiner question quality depends on input material quality:** the MVP clearly restricts input (plain text or clearly legible photographed text) to avoid extraction errors.
- **AI grading of free-form answers may be inconsistent:** the grading rubric is extracted from the original material once, at the Ingest step, limiting the AI from arbitrarily setting its own grading criteria each turn; users are allowed to give feedback on/dispute grading results.
- **Managing multi-turn conversations costs more in API usage than discrete Q&A:** the system limits the maximum number of question-answer turns per concept (e.g., 3 turns) and the number of concepts tested per session, to control cost and response time.
- **The AI may incorrectly extract prerequisite relationships between concepts:** users must confirm/edit the graph before starting to study; the system does not fully trust AI-generated results automatically.

- **The concept graph may contain cycles** (A is a prerequisite of B, B is a prerequisite of A): the system checks the acyclic (DAG) property when saving, rejecting or flagging edges that cause cycles.
- **Some materials/subjects lack a clear layered structure**, making the graph too sparse: in these cases, the system applies the ordinary spaced-repetition mechanism for that concept, without forcing a prerequisite relationship to exist.
- **Schedule risk in the Concept Graph Engine** (the project's central algorithm): if Sprint 3 encounters difficulties, the minimum demo criterion is that the graph saves correctly and the trace-back algorithm runs correctly with mock data; the Dashboard visualization portion can be pushed to Sprint 5 without affecting the main technical argument.
- **AI Examiner (Sprint 4) and system completion (Sprint 5) are adjacent with no buffer sprint**: priority is given to stabilizing the Interview feature before expanding scope; Voice Input can be excluded from the MVP scope entirely if more time is needed.
- **Voice Input (if implemented) risks Vietnamese speech recognition errors** on technical terminology, which could cause the AI to incorrectly grade a correct answer: not included in MVP criteria, only trialed at a later stage if time permits.
- **Risk of dependency on an external AI service (Gemini API)**: design an abstraction layer (AI service layer) to allow switching providers if needed; cache results and limit requests per user per day to control rate limits.